

## **DBMS – SELLER AND MANAGEMENT**

### **PROJECT TITTLE – SAVANA PRODUCTS E-COMMERCE ORDER MANAGEMENT SYSTEM**

#### **1.DATABASE DESIGN**

##### **1.1 SELLER TABLE**

| <b>Attribute</b> | <b>Data Type</b> | <b>Key</b>  |
|------------------|------------------|-------------|
| Seller_ID        | Number(5)        | Primary Key |
| Seller_Name      | Varchar2(100)    | Not Null,   |
| Seller_Email     | Varchar2(100)    | Unique      |
| Contact_Number   | Varchar2(15)     | Unique      |
| Business_Type    | Varchar2(50)     | Not Null,   |

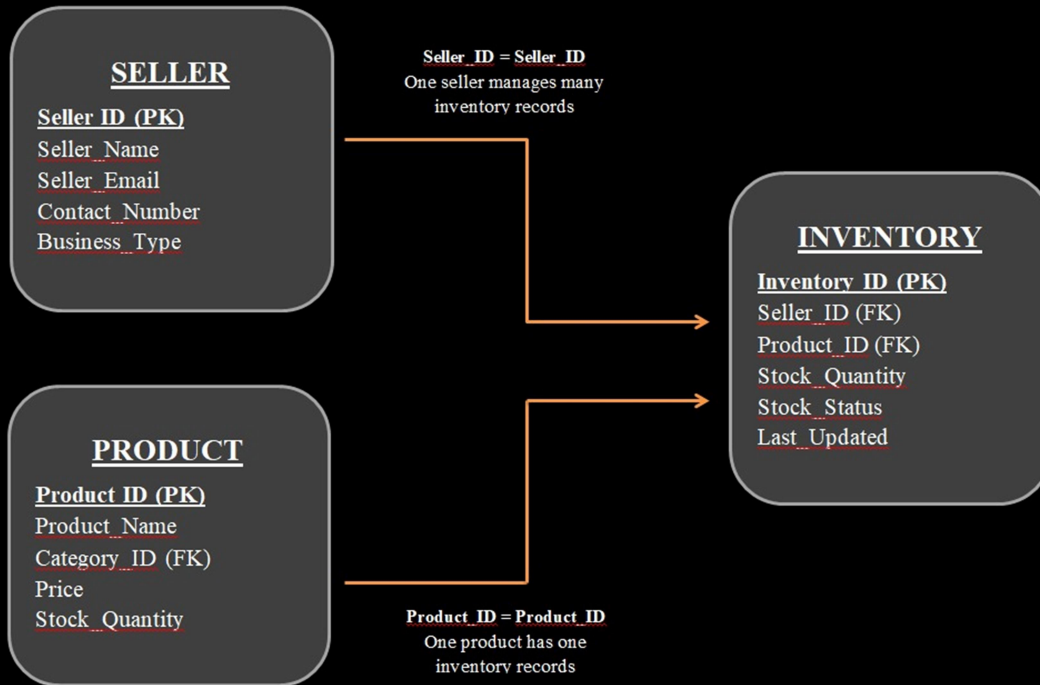
##### **1.2 INVENTORY TABLE**

| <b>Attribute</b> | <b>Data Type</b> | <b>Constraint</b> |
|------------------|------------------|-------------------|
| Inventory_ID     | Number(5)        | Primary Key       |
| Product_ID       | Number(5)        | Foreign Key       |
| Seller_ID        | Number(5)        | Foreign Key       |
| Stock_Quantity   | Number(5)        | NOT NULL          |
| Stock_Status     | Varchar2(20)     | NOT NULL          |
| Last_Updated     | DATE             | NOT NULL          |

## 2.TABLE RELATIONSHIP

### Table Relationship :

The Seller, Product and Inventory tables are connected through primary key and foreign key relationships.



## 3.TABLE CREATION

### 3.1 CREATE SELLER TABLE

```
SQL> CREATE TABLE SHOP_SELLER (  
2     SELLER_ID NUMBER(5) PRIMARY KEY,  
3     SELLER_NAME VARCHAR2(100) NOT NULL,  
4     SELLER_EMAIL VARCHAR2(100) UNIQUE,  
5     CONTACT_NUMBER VARCHAR2(15) NOT NULL UNIQUE,  
6     BUSINESS_TYPE VARCHAR2(50) NOT NULL  
7 );
```

Table created.

## 3.2 CREATE INVENTORY TABLE

```
SQL> CREATE TABLE SHOP_INVENTORY (  
 2     INVENTORY_ID NUMBER(5) PRIMARY KEY,  
 3     SELLER_ID NUMBER(5) NOT NULL,  
 4     PRODUCT_ID NUMBER(5) NOT NULL UNIQUE,  
 5     STOCK_QUANTITY NUMBER(5) NOT NULL,  
 6     STOCK_STATUS VARCHAR2(20) NOT NULL,  
 7     LAST_UPDATED DATE NOT NULL,  
 8     CONSTRAINT FK_INV_SELLER  
 9         FOREIGN KEY (SELLER_ID)  
10         REFERENCES SHOP_SELLER(SELLER_ID),  
11     CONSTRAINT FK_INV_PRODUCT  
12         FOREIGN KEY (PRODUCT_ID)  
13         REFERENCES SHOP_PRODUCT(PRODUCT_ID)  
14 );
```

Table created.

## 4.SELLER DATA INSERTION

```
SQL> INSERT INTO SHOP_SELLER  
 2 VALUES (203, 'Home Needs', 'homeneeds@gmail.com', '9876543212', 'Home Appliances');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO SHOP_SELLER  
 2 VALUES (204, 'Style Hub', 'stylehub@gmail.com', '9876543213', 'Fashion');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO SHOP_SELLER  
 2 VALUES (205, 'Beauty Point', 'beautypoint@gmail.com', '9876543214', 'Beauty');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO SHOP_SELLER  
 2 VALUES (206, 'Smart Zone', 'smartzone@gmail.com', '9876543215', 'Electronics');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO SHOP_SELLER  
 2 VALUES (207, 'Daily Essentials', 'daily@gmail.com', '9876543216', 'Lifestyle');  
  
1 row created.  
  
SQL>  
SQL> COMMIT;  
  
Commit complete.
```

## 5.INVENTORY DATA INSERTION

```
SQL>
SQL> INSERT INTO SHOP_INVENTORY
  2  VALUES (307, 201, 107, 35, 'AVAILABLE', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO SHOP_INVENTORY
  2  VALUES (308, 205, 108, 15, 'AVAILABLE', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO SHOP_INVENTORY
  2  VALUES (309, 204, 109, 0, 'UNAVAILABLE', SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO SHOP_INVENTORY
  2  VALUES (310, 203, 110, 18, 'AVAILABLE', SYSDATE);

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

## 6.SELLER PRODUCT INFORMATION

### 6.1 DISPLAY SELLER AND PRODUCT DETAILS

```
SQL> SET LINESIZE 130
SQL>
SQL> COLUMN SELLER_NAME FORMAT A20
SQL> COLUMN PRODUCT_NAME FORMAT A25
SQL> COLUMN STOCK_STATUS FORMAT A15
SQL>
SQL> SELECT
  2     s.SELLER_ID,
  3     s.SELLER_NAME,
  4     p.PRODUCT_ID,
  5     p.PRODUCT_NAME,
  6     i.STOCK_QUANTITY,
  7     i.STOCK_STATUS
  8 FROM SHOP_SELLER s
  9 JOIN SHOP_INVENTORY i
10 ON s.SELLER_ID = i.SELLER_ID
11 JOIN SHOP_PRODUCT p
12 ON i.PRODUCT_ID = p.PRODUCT_ID
13 ORDER BY s.SELLER_ID;
```

| SELLER_ID | SELLER_NAME  | PRODUCT_ID | PRODUCT_NAME        | STOCK_QUANTITY | STOCK_STATUS |
|-----------|--------------|------------|---------------------|----------------|--------------|
| 201       | Trendy Store | 107        | Casual Shirt        | 35             | AVAILABLE    |
| 201       | Trendy Store | 103        | Cotton Shirt        | 40             | AVAILABLE    |
| 202       | Tech World   | 101        | Wireless Headphones | 30             | AVAILABLE    |
| 202       | Tech World   | 102        | Smart Watch         | 25             | AVAILABLE    |
| 203       | Home Needs   | 105        | Mixer Grinder       | 0              | UNAVAILABLE  |
| 203       | Home Needs   | 110        | Electric Kettle     | 18             | AVAILABLE    |
| 204       | Style Hub    | 109        | Handbag             | 0              | UNAVAILABLE  |
| 204       | Style Hub    | 104        | Denim Jeans         | 30             | AVAILABLE    |
| 205       | Beauty Point | 108        | Face Wash           | 15             | AVAILABLE    |
| 206       | Smart Zone   | 106        | Bluetooth Speaker   | 20             | AVAILABLE    |

10 rows selected.

## 7.INVENTORY STATUS

### 7.1 DISPLAY AVAILABLE PRODUCTS

```
SQL> SET LINESIZE 100
SQL>
SQL> COLUMN PRODUCT_NAME FORMAT A25
SQL>
SQL> SELECT
  2     p.PRODUCT_ID,
  3     p.PRODUCT_NAME,
  4     i.STOCK_QUANTITY,
  5     'AVAILABLE' AS STOCK_STATUS
  6 FROM SHOP_PRODUCT p
  7 JOIN SHOP_INVENTORY i
  8 ON p.PRODUCT_ID = i.PRODUCT_ID
  9 WHERE i.STOCK_QUANTITY > 0;
```

| PRODUCT_ID | PRODUCT_NAME        | STOCK_QUANTITY | STOCK_STATUS |
|------------|---------------------|----------------|--------------|
| 101        | Wireless Headphones | 30             | AVAILABLE    |
| 106        | Bluetooth Speaker   | 20             | AVAILABLE    |
| 107        | Casual Shirt        | 35             | AVAILABLE    |
| 108        | Face Wash           | 15             | AVAILABLE    |
| 102        | Smart Watch         | 25             | AVAILABLE    |
| 103        | Cotton Shirt        | 40             | AVAILABLE    |
| 104        | Denim Jeans         | 30             | AVAILABLE    |
| 110        | Electric Kettle     | 18             | AVAILABLE    |

8 rows selected.

## 7.2 DISPLAY UNAVAILABLE PRODUCTS

```
SQL> SET LINESIZE 100
SQL>
SQL> COLUMN PRODUCT_NAME FORMAT A25
SQL>
SQL> SELECT
  2     p.PRODUCT_ID,
  3     p.PRODUCT_NAME,
  4     i.STOCK_QUANTITY,
  5     'AVAILABLE' AS STOCK_STATUS
  6 FROM SHOP_PRODUCT p
  7 JOIN SHOP_INVENTORY i
  8 ON p.PRODUCT_ID = i.PRODUCT_ID
  9 WHERE i.STOCK_QUANTITY > 0;
```

| PRODUCT_ID | PRODUCT_NAME        | STOCK_QUANTITY | STOCK_STATUS |
|------------|---------------------|----------------|--------------|
| 101        | Wireless Headphones | 30             | AVAILABLE    |
| 106        | Bluetooth Speaker   | 20             | AVAILABLE    |
| 107        | Casual Shirt        | 35             | AVAILABLE    |
| 108        | Face Wash           | 15             | AVAILABLE    |
| 102        | Smart Watch         | 25             | AVAILABLE    |
| 103        | Cotton Shirt        | 40             | AVAILABLE    |
| 104        | Denim Jeans         | 30             | AVAILABLE    |
| 110        | Electric Kettle     | 18             | AVAILABLE    |

8 rows selected.

## 8. INVENTORY UPDATE

### DEMONSTRATE UPDATING STOCK INFORMATION

```
SQL> UPDATE SHOP_INVENTORY
  2 SET STOCK_QUANTITY = 20,
  3     STOCK_STATUS = 'AVAILABLE',
  4     LAST_UPDATED = SYSDATE
  5 WHERE INVENTORY_ID = 305;
```

1 row updated.

```
SQL>
SQL> COMMIT;
```

Commit complete.

## 9. INVENTORY STATUS REPORTS

### 9.1 COMPLETE INVENTORY REPORT

```
SQL> SET LINESIZE 140
SQL>
SQL> COLUMN SELLER_NAME FORMAT A20
SQL> COLUMN PRODUCT_NAME FORMAT A25
SQL> COLUMN STOCK_STATUS FORMAT A15
SQL>
SQL> SELECT
  2     p.PRODUCT_ID,
  3     p.PRODUCT_NAME,
  4     s.SELLER_NAME,
  5     i.STOCK_QUANTITY,
  6     i.STOCK_STATUS,
  7     i.LAST_UPDATED
  8 FROM SHOP_PRODUCT p
  9 JOIN SHOP_INVENTORY i
10 ON p.PRODUCT_ID = i.PRODUCT_ID
11 JOIN SHOP_SELLER s
12 ON i.SELLER_ID = s.SELLER_ID
13 ORDER BY p.PRODUCT_ID;
```

| PRODUCT_ID | PRODUCT_NAME        | SELLER_NAME  | STOCK_QUANTITY | STOCK_STATUS | LAST_UPDA |
|------------|---------------------|--------------|----------------|--------------|-----------|
| 101        | Wireless Headphones | Tech World   | 30             | AVAILABLE    | 17-SEP-26 |
| 102        | Smart Watch         | Tech World   | 25             | AVAILABLE    | 17-SEP-26 |
| 103        | Cotton Shirt        | Trendy Store | 40             | AVAILABLE    | 17-SEP-26 |
| 104        | Denim Jeans         | Style Hub    | 30             | AVAILABLE    | 17-SEP-26 |
| 105        | Mixer Grinder       | Home Needs   | 20             | AVAILABLE    | 17-SEP-26 |
| 106        | Bluetooth Speaker   | Smart Zone   | 20             | AVAILABLE    | 17-SEP-26 |
| 107        | Casual Shirt        | Trendy Store | 35             | AVAILABLE    | 17-SEP-26 |
| 108        | Face Wash           | Beauty Point | 15             | AVAILABLE    | 17-SEP-26 |
| 109        | Handbag             | Style Hub    | 0              | UNAVAILABLE  | 17-SEP-26 |
| 110        | Electric Kettle     | Home Needs   | 18             | AVAILABLE    | 17-SEP-26 |

10 rows selected.

### 9.2 AVAILABLE AND UNAVAILABLE PRODUCT COUNT

```
SQL> SELECT
  2     STOCK_STATUS,
  3     COUNT(*) AS PRODUCT_COUNT
  4 FROM SHOP_INVENTORY
  5 GROUP BY STOCK_STATUS;
```

| STOCK_STATUS | PRODUCT_COUNT |
|--------------|---------------|
| AVAILABLE    | 9             |
| UNAVAILABLE  | 1             |

### 9.3 TOTAL STOCK BY SELLER

```
SQL> SELECT
  2     s.SELLER_ID,
  3     s.SELLER_NAME,
  4     SUM(i.STOCK_QUANTITY) AS TOTAL_STOCK
  5 FROM SHOP_SELLER s
  6 JOIN SHOP_INVENTORY i
  7 ON s.SELLER_ID = i.SELLER_ID
  8 GROUP BY s.SELLER_ID, s.SELLER_NAME
  9 ORDER BY s.SELLER_ID;
```

| SELLER_ID | SELLER_NAME  | TOTAL_STOCK |
|-----------|--------------|-------------|
| 201       | Trendy Store | 75          |
| 202       | Tech World   | 55          |
| 203       | Home Needs   | 38          |
| 204       | Style Hub    | 30          |
| 205       | Beauty Point | 15          |
| 206       | Smart Zone   | 20          |

6 rows selected.

### 9.4 PRODUCTS WITH ZERO STOCK

**THIS IS PARTICULARLY USEFUL FOR DEMONSTRATING  
UNAVAILABLE PRODUCTS**

```
SQL> SELECT
  2     p.PRODUCT_ID,
  3     p.PRODUCT_NAME,
  4     s.SELLER_NAME,
  5     i.STOCK_QUANTITY
  6 FROM SHOP_PRODUCT p
  7 JOIN SHOP_INVENTORY i
  8 ON p.PRODUCT_ID = i.PRODUCT_ID
  9 JOIN SHOP_SELLER s
 10 ON i.SELLER_ID = s.SELLER_ID
 11 WHERE i.STOCK_QUANTITY = 0;
```

| PRODUCT_ID | PRODUCT_NAME | SELLER_NAME | STOCK_QUANTITY |
|------------|--------------|-------------|----------------|
| 109        | Handbag      | Style Hub   | 0              |